Hierarchical Latent Exchange & Coordination Fabric (HLEX)

Systems Design Proposal for xAI Research & Training Infrastructure

A detailed technical proposal for a closed, geometry-aware multi-model coordination layer
that enables real-time exchange of rare-token statistics, embedding sketches, and higher-order latent structure.

Prepared for consideration by the xAI Research & Training teams
Document version 1.1 (expanded) | August 2026
Author: Benjamin Frohman

0. Author Research Background

This proposal is grounded in independent research at the intersection of geometric analysis, topological methods, and hierarchical representation learning. The author's ongoing work focuses on the geometric structure of high-dimensional solution spaces and the construction of hierarchical, formally grounded frameworks that can detect and organize novel structure—precisely the concerns that motivate HLEX.

Primary research threads relevant to the present design include:

- **Geometric and symplectic approaches to regularity questions in nonlinear PDEs**, with particular attention to the structure of singularity formation and the organization of solution manifolds. This work has produced detailed conjectural frameworks and extensive Lean 4 formalization efforts aimed at making geometric claims machine-checkable.
- **Hierarchical and dual representations** (symplectic/holographic duals, tether-style constructions) that attempt to reorganize the analysis of complex dynamical systems into more tractable geometric objects. The same impulse—finding a higher-order geometric language that reveals structure invisible at the base level—underpins the second-order modeling layer of HLEX.
- **Topological and geometric descriptors of high-dimensional data**, including persistent homology, intrinsic dimension estimates, and manifold-aware summaries. These tools appear in the present design as candidate geometric classifications that the Ontology & Geometry Registry can attach to latent regions.
- **Formal verification and Lean 4 infrastructure** for large-scale mathematical arguments. Experience building and maintaining substantial formal libraries informs the design emphasis on versioned, auditable, and researcher-inspectable structure in the Registry and policy layers.

The author is currently completing undergraduate studies with the intention of teaching mathematics while continuing independent research. The HLEX proposal is offered both as a concrete systems contribution and as a demonstration of the ability to translate geometric and topological insight into operational infrastructure that frontier AI teams can actually build and measure.

Contact and further materials (including formalization repositories and technical notes on geometric descriptors) are available upon request.

1. Executive Summary

This document presents a complete systems design for the Hierarchical Latent Exchange & Coordination Fabric (HLEX). HLEX is a practical infrastructure layer that allows multiple models (or successive generations of the same model family) under unified organizational control to exchange structured signals about novel structure appearing in their latent spaces. These signals consist of rare-token / novelty markers and compact embedding sketches. Concurrent appearance of the same markers across models is treated as a first-class training and routing signal. The system further supports the training of explicit second-order models that operate on the space of latent representations themselves, together with a researcher-facing Ontology & Geometry Registry that turns statistical co-occurrences into named, scientifically interpretable objects.

The design is deliberately incremental. Every phase produces usable artifacts even if later phases are deferred. Marker extraction overhead is engineered to remain well under 1 % of training compute. The architecture reuses existing logging, experiment-tracking, and message-bus infrastructure where possible, minimizing new surface area while creating capabilities that are currently absent from standard large-model training stacks.

HLEX directly advances three research directions of high relevance to frontier AI work: (1) the geometric characterization of latent spaces, (2) the construction of formal ontological structure over high-dimensional representations, and (3) hierarchical / multi-model coordination that treats co-occurrence of novel markers as an adaptive signal. The resulting system is intended both as production infrastructure and as a research instrument that lets scientists formulate and test geometric and ontological hypotheses with far lower iteration cost than full model retraining.

2. Motivation & Strategic Alignment

Contemporary large-model training still treats each training run as largely independent. Shared knowledge is limited to static datasets, occasional distillation, and post-hoc contamination audits. When a model encounters genuinely novel structure—patterns that lie far from its current training manifold—the signal is usually local to that run and is not systematically propagated to sibling or successor models. The result is duplicated discovery effort, slower curriculum adaptation, and missed opportunities for coordinated multi-agent behavior.

xAI's stated mission—to understand the true nature of the universe—requires models that can efficiently locate and exploit novel structure rather than repeatedly rediscovering already-known regions of representation space. A coordination fabric that makes concurrent novelty visible across the fleet, and that can convert those signals into concrete training and routing actions, is a direct infrastructural contribution to that mission. In addition, the second-order modeling layer and the Ontology & Geometry Registry create a new experimental surface on which researchers can study the geometry of latent spaces and impose ontological constraints at a level of abstraction that is currently difficult to reach.

The design therefore serves two simultaneous goals: (a) measurable improvement in training efficiency, synthetic-data quality, and multi-model coordination; and (b) provision of a living laboratory for geometric and ontological research on the representations that frontier models actually use.

3. Design Goals & Non-Goals

Primary Goals

- Enable low-overhead, continuous extraction of rare-token novelty markers and compact embedding sketches from participating training and inference processes.
- Provide a durable, multi-tenant signal bus that aggregates co-occurrence statistics across models and makes them available at multiple time scales.
- Support the training and serving of explicit second-order models whose inputs are solely the latent-derived signals of first-order models.
- Allow researchers to define policies that convert co-occurrence patterns into concrete actions (routing, synthetic-data prioritization, curriculum adjustment, alerts).
- Maintain a versioned Ontology & Geometry Registry that attaches human- and machine-readable structure to latent regions and marker clusters.
- Keep marker-extraction overhead under 1 % of training FLOPs and design every component for incremental adoption.

Explicit Non-Goals (v1)

- Cross-organization model exchange or open federation. HLEX assumes a closed ecosystem under unified administrative control.
- Replacement of existing experiment-tracking or model-versioning systems; HLEX is designed to interoperate with them.
- Fully automatic scientific discovery without human oversight of major policy changes and ontological commitments.
- Support for models outside the controlled fleet (no public watermarking or external triangulation).

4. High-Level Architecture

HLEX is organized into five cooperating layers. The layers communicate through well-defined, versioned interfaces so that each can evolve independently. The overall data flow is unidirectional from first-order models toward higher-order structure, with controlled feedback paths for actions.

4.1 Layer Overview

1. Marker Extraction Layer — runs inside or adjacent to every participating training job and selected inference endpoints. Continuously (or at configurable intervals) emits rare-token statistics and embedding sketches.

2. Shared Latent Signal Bus — a high-throughput, durable, multi-tenant event bus that receives raw signals, performs time-windowed aggregation and cross-model joins, and materializes feature stores for downstream consumers.

3. Second-Order Model Service — trains and serves models whose sole inputs are marker time series, embedding sketches, and co-occurrence graphs produced by the first-order fleet.

4. Feedback & Routing Controller — evaluates researcher-defined policies against second-order outputs and raw co-occurrence events, then emits concrete actions into data pipelines, routing tables, or job schedulers.

5. Ontology & Geometry Registry — a versioned knowledge base that associates names, geometric classifications, and ontological constraints with latent regions and marker clusters, making statistical signals scientifically legible.

4.2 Conceptual Data Flow

First-order model → Marker Extraction → Signal Bus → (aggregation + feature store) → Second-Order Models → Feedback Controller → actions back into training / routing / synthetic-data systems. In parallel, researchers and automated clustering processes write into and read from the Ontology & Geometry Registry, enriching future signal interpretation.

5. Marker Extraction Layer — Detailed Specification

The Marker Extraction Layer is the only component that must touch the critical path of training or high-volume inference. Consequently it is designed for extreme efficiency and configurability.

5.1 Novelty Marker Definition

A novelty marker is a token, short n-gram ($n \leq 4$), or hashed feature whose observed frequency in a recent window falls below a dynamic threshold relative to a reference distribution. The reference may be a static reference corpus, an exponentially-weighted moving average of the model's own recent token statistics, or a fleet-wide reference maintained by the Signal Bus.

Markers are emitted only when their rarity score exceeds a configurable threshold. This keeps the event rate manageable even for very large vocabularies.

5.2 Embedding Sketches

Full intermediate activations are far too large to stream continuously. Instead the extractor produces compact sketches of selected layers. Recommended methods, ordered by increasing fidelity/cost:

- **Count-Min Sketch or Feature Hashing** of activation magnitudes — extremely cheap, good for coarse distributional monitoring.
- **Random projections (Johnson-Lindenstrauss)** of pooled or CLS-style vectors — preserves approximate distances.
- **Quantile sketches or t-digest** of selected activation coordinates — useful for detecting distributional shift.
- **Low-rank or sparse autoencoder projections** when a pre-trained sketching model is available.

5.3 Emission Protocol (schema sketch)

```
message MarkerEvent {
  string model_id = 1;          // lineage / experiment id
  string run_id = 2;
  int64  step = 3;
  int64  timestamp_ns = 4;
  repeated TokenMarker markers = 5;
  repeated EmbeddingSketch sketches = 6;
  map<string, string> tags = 7;
}
message TokenMarker {
  string token_or_ngram = 1;
  float  rarity_score = 2;
  uint64 count_in_window = 3;
}
message EmbeddingSketch {
  string layer_name = 1;
  string sketch_type = 2;       // "jl", "countmin", "quantile", ...
  bytes  payload = 3;
  uint32 dimension = 4;
}
```

Emission cadence is configurable: every N training steps, every M seconds of wall time, or on a statistical trigger. Inference endpoints may use reservoir sampling. Target overhead: total additional FLOPs and memory bandwidth < 1 % of the host model's forward/backward pass.

6. Shared Latent Signal Bus

The bus is the central nervous system of HLEX. It must be durable, multi-tenant, ordered, and capable of both real-time fan-out and efficient batch materialization.

6.1 Topic & Tenancy Model

```
hlex.v1.markers.raw.<project>.<model_family>
hlex.v1.sketches.raw.<project>.<model_family>
hlex.v1.agg.cooccur.<window>.<project>
hlex.v1.features.second_order.<project>
```

Access control is enforced at the topic level and keyed to model lineage and research-project identity. Aggregated and anonymized views may be shared more widely under policy.

6.2 Aggregation Services

Streaming aggregation jobs maintain sliding-window co-occurrence counts. Recommended windows: 1 min, 15 min, 1 h, 6 h, 24 h, and all-time (with exponential decay). Cross-model joins are performed by marker identity. When the same marker appears in two or more model lineages inside a window, a co-occurrence event is recorded together with participating model IDs and rarity scores.

6.3 Retention & Down-sampling

Raw high-frequency events are retained for a short period (hours to a few days). Aggregated co-occurrence matrices and sketch summaries are retained far longer with automatic progressive down-sampling, keeping storage costs bounded while preserving long-term geometric signal.

7. Second-Order Model Service

Second-order models never see raw text or original training tokens. Their entire input distribution consists of the structured signals produced by the first-order fleet. This creates a clean separation of concerns and opens a new research surface—the geometry and dynamics of the space of latent spaces.

7.1 Input Representations

- Time series of marker rarity scores and hit counts, optionally grouped by ontological category.
- Sequences or bags of embedding sketches, aligned across models by time or training step.
- Dynamic graphs whose nodes are markers or latent regions and whose edges are co-occurrence strengths.
- Derived features: novelty density, cross-model agreement scores, drift magnitudes.

7.2 Model-Class Suggestions

- **Sequence models** (Transformer or state-space) over marker time series — forecasting novelty bursts.
- **Graph neural networks** on co-occurrence graphs — community detection and region labeling.
- **Autoencoders / variational models** on sketch collections — learning a continuous map of fleet latent geometry.
- **Simple statistical or boosting models** as strong baselines and for low-latency decisions.

7.3 Downstream Uses

- Predict regions of latent space becoming active across the fleet.
- Detect coordinated novelty invisible to any single model.
- Propose synthetic-data generation targets or curriculum re-weightings.
- Serve as a learned component inside multi-agent or MoE routers.
- Provide early-warning signals of distributional shift or data-quality problems.

8. Feedback & Routing Controller

The Controller is the only component allowed to emit actions back into the training and serving infrastructure. It evaluates researcher-defined policies against the current state of the Signal Bus and the latest second-order model outputs.

8.1 Policy Interface (example)

```
policy novelty_surge_alert:
  when cooccurrence_count(marker, window=1h) >= 3
   and mean_rarity(marker) > 0.92
   and marker not in Registry.known_benign
  then emit Alert(severity=high, marker=marker, models=participants)

policy synthetic_data_boost:
  when second_order.predict_novelty_density() > threshold
  then increase_sampling_weight(category=predicted_region, factor=1.8)
```

All actions pass through a safety and resource-quota layer. Major policy classes can be placed behind human approval gates during early deployment.

8.2 Initial Action Types

- Adjust sampling weights or generation prompts for synthetic-data pipelines.
- Modify expert routing probabilities or gating thresholds in MoE models.
- Raise research alerts (internal dashboard / ticket).
- Trigger targeted evaluation suites or fine-tuning jobs.
- Update entries or confidence scores in the Ontology & Geometry Registry (under policy).

9. Ontology & Geometry Registry

Statistical co-occurrence alone is not science. The Registry supplies the missing semantic and geometric layer that turns raw markers into research objects.

9.1 Core Entities

- **Marker** — the atomic observed unit (token, n-gram, or sketch-hash).
- **Latent Region** — a named cluster or manifold region in embedding space, possibly defined by a geometric descriptor.
- **Ontological Concept** — a higher-level label or relation that researchers attach to regions or markers.
- **Geometric Classification** — explicit geometric attributes (approximate dimension, curvature estimates, topological features such as persistent homology barcodes, distance-to-manifold statistics, etc.).
- **Provenance & Confidence** — which models observed the marker, when, under what training conditions, and how confident the association is.

9.2 Intended Workflow

Researchers (or automated clustering jobs) propose new Latent Regions or link existing markers to Ontological Concepts. The Registry stores these assertions with full versioning and audit history. Subsequent co-occurrence events can be automatically enriched with the current ontological labels, making second-order models and human dashboards far more interpretable. Over time the Registry becomes a living map of the geometric and conceptual structure that the model fleet has discovered.

10. Implementation Roadmap (Summary)

The roadmap is staged so that each phase delivers independent value. A fully detailed Phase 0 breakdown appears in Appendix A.

Phase 0 — Foundation (2–4 weeks)

- Schemas, offline instrumentation of 1–2 runs, batch co-occurrence, overhead measurement, design review.

Phase 1 — Bus & Minimal Feedback (4–6 weeks)

- Deploy Signal Bus, streaming aggregators, minimal Feedback Controller, researcher dashboards, initial Registry entries.

Phase 2 — Second-Order Models (6–10 weeks)

• Training/serving pipelines for sequence + graph second-order models, integration under feature flags, expanded Registry, controlled experiments.

Phase 3 — Production Hardening & Geometric Depth (ongoing)

• Real-time paths, stronger isolation, deeper geometric tooling (including persistent homology), multi-agent routing experiments.

11. Success Metrics & Evaluation

Infrastructural Metrics

• Marker extraction overhead (FLOPs, memory bandwidth, wall-time) remains < 1 %.
• End-to-end latency from emission to aggregated co-occurrence view meets design targets for each window.
• Bus availability, durability, and multi-tenant isolation meet internal SLOs.

Utility Metrics

• Improvement in synthetic-data quality proxies attributable to novelty-boosted data.
• Reduction in redundant discovery of the same novel regions across successive training runs.
• Researcher adoption: number of active policies, Registry entries, and dashboard sessions.
• Qualitative feedback on whether geometric and ontological views accelerate hypothesis formation.

12. Risks, Security & Mitigations

Risk	Impact	Mitigation
Bandwidth / storage explosion from high-frequency markers	High	Aggressive sketching, configurable sampling, progressive down-sampling, hard rate limits
Noisy or spurious co-occurrences driving bad actions	Medium–High	Statistical thresholds, second-order filtering, human approval gates, Registry allow/deny lists
Cross-project information leakage	High	Strict topic-level ACLs, optional differential-privacy noise on sketches, full audit logging
Second-order models over-fitting to transient artifacts	Medium	Held-out evaluation, regular re-training discipline, human review of major policy promotions
Engineering complexity slowing adoption	Medium	Strict phased delivery; every phase ships independently usable value; reuse of existing infrastructure

Security posture follows least privilege. Raw marker streams are treated as sensitive research data. Second-order models and the Registry are additional attack surfaces and must be included in ordinary model-security and data-governance processes.

13. Future Extensions & Research Directions

Once the core fabric is operational, several high-leverage extensions become natural:

• **Richer geometric descriptors** — persistent homology summaries, local intrinsic dimension estimates, approximate geodesic distances, and simple Riemannian metrics attached to Latent Regions (see Appendix C for a deeper treatment of persistent homology).
• **Ontological reasoning layer** — lightweight inference over the Registry graph so that co-occurrence of markers can trigger higher-level conceptual hypotheses.
• **Cross-generational transfer** — using second-order models trained on one model family to accelerate or regularize the training of a successor family.
• **Interactive geometric exploration tools** — researcher UIs that let scientists navigate the living map of latent regions and directly propose new markers or policies.
• **Integration with formal verification efforts** — where geometric or ontological claims about latent structure can be linked to more rigorous, machine-checkable statements.

14. Conclusion & Offer

HLEX supplies a missing coordination layer for large-scale model training: a closed, geometry-aware fabric that makes concurrent novelty visible, supports explicit second-order modeling of the space of latent representations, and turns statistical co-occurrence into scientifically interpretable structure through an Ontology & Geometry Registry. The design is incremental, overhead-conscious, and deliberately aligned with the research and infrastructure realities of a frontier AI organization.

The present document is offered as a concrete technical proposal and as evidence of systems-level thinking about latent geometry, multi-model coordination, and the infrastructural prerequisites for more systematic scientific discovery with large models. I would welcome the opportunity to discuss the design in detail, refine it against xAI's actual constraints and priorities, and contribute to its realization—whether as a research collaborator, an infrastructure engineer focused on the latent-coordination problem, or in any other capacity the team finds useful.

Benjamin Frohman
Independent researcher — geometric analysis, hierarchical representation, topological methods, and multi-model systems

Appendix A — Detailed Phase 0 Implementation Plan

Phase 0 is deliberately scoped to produce measurable evidence of feasibility and overhead before any production bus or feedback loop is built. The goal is a short, high-signal design review that gives leadership a clear go / no-go decision for Phase 1.

A.1 Objectives & Success Criteria

- Freeze v0 schemas that can be versioned without breaking later phases.
- Demonstrate offline marker + sketch extraction on at least two real training runs.
- Produce a batch co-occurrence matrix and basic statistics from those runs.
- Publish a measured overhead number (FLOPs / wall-time / memory) on a representative node.
- Deliver a written Phase-0 retrospective and a concrete Phase-1 resource estimate.

Success = all five items completed and reviewed; failure of any one item is grounds for re-scoping before Phase 1.

A.2 Workstream 1 — Schema & Tooling (Week 1)

1. Create a private design repository (or designated design-doc location) and seed it with this proposal + a short README.

2. Draft and freeze v0 protobuf (or Arrow) schemas for MarkerEvent, TokenMarker, and EmbeddingSketch. Include explicit version fields.

3. Generate language bindings (at least Python and C++/the internal training language) and a minimal validation library.

4. Write a one-page “Schema Contract” document that later phases must respect.

Acceptance: schemas reviewed by one researcher + one engineer; bindings compile and round-trip a synthetic event.

A.3 Workstream 2 — Instrumentation (Weeks 1–2)

5. Select two existing training runs that differ in model scale or data mixture (diversity is more valuable than perfection).

6. Add offline logging hooks that, every N steps (configurable, start with N = 50 or 100), compute and write MarkerEvents to a local or shared filesystem path. Prefer non-blocking, asynchronous write.

7. Implement a minimal rarity scorer: exponential moving average of unigram and bigram frequencies is sufficient. Document the exact formula and decay constants.

8. Implement at least one sketching method. Recommended starting points: (a) Count-Min sketch of activation magnitudes on a chosen mid-layer, or (b) a fixed Johnson-Lindenstrauss projection of a pooled residual-stream vector. Keep the implementation under 50 lines of core logic if possible.

Acceptance: both runs produce valid MarkerEvent files; a simple inspection script can print the top-k rarest tokens per run.

A.4 Workstream 3 — Batch Co-occurrence & Analysis (Week 2–3)

9. Write a batch job (Spark, Ray, Dask, or the internal equivalent) that reads the two MarkerEvent streams, joins on token/ngram identity, and materializes a co-occurrence count matrix together with per-marker rarity statistics.

10. Produce a short notebook or report that answers: How many markers are unique to each run? How many are shared? What is the distribution of co-occurrence counts? Are there any qualitatively surprising shared rare tokens?

Acceptance: the matrix and the report exist; a second person can re-run the job from the logged events and obtain identical counts.

A.5 Workstream 4 — Overhead Measurement (Week 3)

11. On a representative training node (same GPU/accelerator type used for production runs), measure wall-time, estimated FLOPs, and peak memory attributable to the extraction hooks versus a baseline run without them.

12. Document the measurement methodology (tools used, number of steps averaged, variance observed) and the resulting overhead percentages.

Acceptance: overhead report exists; if overhead exceeds 1.5 % the team must either optimize the hooks or explicitly justify a higher budget before Phase 1.

A.6 Workstream 5 — Synthesis & Decision (Week 3–4)

13. Write a Phase-0 Retrospective (2–4 pages) covering what worked, what was harder than expected, measured overhead, and any schema changes required.

14. Produce a Phase-1 Proposal with rough headcount, calendar estimate, and risk notes.

15. Present findings to research & training leadership for an explicit go / no-go / re-scope decision on Phase 1.

Acceptance: written materials exist; decision is recorded in the design repository.

A.7 Suggested Team Composition for Phase 0

- 1–2 research scientists (to define interesting markers, interpret co-occurrence results, and keep the geometric/ontological goals visible).
- 1–2 infrastructure / training engineers (to own schemas, hooks, batch jobs, and measurement).
- Optional: one part-time formal-methods or geometry-aware collaborator if the team wants early input on Registry design.

Total effort is intentionally small. Phase 0 is a probe, not a product.

Appendix B — Glossary of Key Terms

Novelty Marker / Rare-Token Marker — A token or short n-gram whose frequency in a recent window is anomalously low relative to a reference distribution; the operational unit of the original ‘egg’ concept.

Embedding Sketch — A compact, lossy summary of selected latent activations designed for continuous streaming and approximate geometric comparison.

Co-occurrence Event — The observation that the same marker (or sufficiently similar sketch) appears in two or more model lineages inside a defined time or step window.

Second-Order Model — A model whose training distribution consists solely of the structured latent-derived signals of first-order models; it learns in the space of latent spaces.

Ontology & Geometry Registry — The versioned store that attaches names, geometric descriptors, and conceptual structure to markers and latent regions, making raw statistics scientifically legible.

Feedback Controller — The policy engine that converts second-order outputs and co-occurrence statistics into concrete actions on data pipelines, routers, or job schedulers.

Appendix C — Persistent Homology Applications for Latent Geometry in HLEX

Persistent homology (PH) is a tool from topological data analysis that quantifies the shape of a point cloud across scales. In the context of HLEX it supplies one of the most natural geometric classifications that the Ontology & Geometry Registry can attach to Latent Regions. This appendix sketches how PH can be made operational inside the fabric.

C.1 Why Persistent Homology?

Latent activations of large models form extremely high-dimensional point clouds. Classical clustering or linear subspace methods capture only part of their structure. Persistent homology tracks the birth and death of topological features (connected components, loops, voids, ...) as a scale parameter varies. The resulting persistence diagrams or barcodes are stable under small perturbations and give a multi-scale summary that is complementary to distances, densities, and intrinsic-dimension estimates.

For HLEX the practical value is that a Latent Region can be annotated not only with a name and an ontological label but also with a compact topological signature: “this region contains a robust 1-dimensional hole that persists over a wide range of scales,” or “the 0-dimensional persistence suggests three well-separated sub-clusters.” Such signatures can later be used by second-order models or by human researchers as features.

C.2 Concrete Integration Points

- **Sketch enrichment** — When embedding sketches are materialised into the feature store, an optional offline (or low-frequency online) PH pipeline can be run on sampled batches of sketches belonging to the same model lineage or the same co-occurrence community. The resulting persistence diagrams are stored as additional geometric descriptors on the corresponding Latent Region.
- **Registry geometric classification** — The Geometry Registry entity already contemplates “topological features.” Persistence barcodes or their vectorisations (persistence images, landscapes, or simple birth/death statistics) become first-class attributes that researchers can query and that second-order models can consume.
- **Novelty characterisation** — A newly appearing marker cluster whose persistence diagram differs markedly from previously observed diagrams can be flagged as topologically novel, even if its raw token rarity is only moderate. This supplies an orthogonal novelty signal.
- **Cross-model comparison** — Persistence diagrams of corresponding regions across two model lineages can be compared with the bottleneck or Wasserstein distance. Large distances become another co-occurrence / divergence feature for the Feedback Controller.

C.3 Practical Computational Path

Full PH on millions of high-dimensional points is expensive. The recommended path inside HLEX is:

1. Sub-sample or sketch the activation points aggressively (the same sketches already flowing through the bus are a natural starting point).
2. Optionally project to a moderate dimension (e.g., 32–128) with a data-dependent or random projection that preserves the features of interest.
3. Compute PH with a modern library (Ripser, giotto-tda, Gudhi, or an internal equivalent) using a sparse filtration or approximate nearest-neighbour graph.
4. Vectorise the diagrams into a fixed-length descriptor that can be stored and compared cheaply.
5. Attach the descriptor to the Latent Region in the Registry and expose it to second-order training.

Because PH is run offline or at low frequency on already-reduced sketches, it does not disturb the critical-path overhead budget of marker extraction.

C.4 Research Questions Opened by the Integration

- Do topologically robust regions in latent space correspond to more stable or more transferable features across model generations?
- Can persistence-based novelty scores improve synthetic-data prioritisation beyond pure rarity?
- Is there a useful relationship between the persistence of a region and the difficulty that first-order models have in representing the corresponding concepts?
- How do persistence diagrams evolve across successive training runs of the same model family, and can second-order models learn to predict that evolution?

These questions are deliberately left open; HLEX’s role is to make the geometric measurements cheap and systematic enough that the research team can answer them with data rather than with isolated case studies.

C.5 Relation to the Author’s Broader Geometric Interests

The author’s independent work on geometric and symplectic structures in nonlinear PDE solution spaces has repeatedly encountered the same underlying problem: high-dimensional dynamical objects possess organisation that is invisible to local analytic estimates yet becomes visible once the right multi-scale geometric or topological language is applied. Persistent homology is one mature language of that kind. Embedding it inside a production coordination fabric such as HLEX is a natural way to test whether the same geometric intuition that has proved useful in continuum analysis can also organise the discrete, high-dimensional latent spaces of large models.

End of Document — Hierarchical Latent Exchange & Coordination Fabric (HLEX) v1.1

Prepared for xAI Research & Training consideration | Expanded edition with Phase-0 detail, Persistent Homology appendix, and Author Research Background